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Continual Instruction Tuning via Increasing Experts

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***Abstract*—Large language models (LLMs) and multimodal models (MMs) have exhibited impressive capabilities in various domains, particularly in general language understanding and visual reasoning. However, these models, trained on massive data, may not be finely optimized for specific tasks triggered by instructions. Continual instruction tuning is crucial to adapting a large model to evolving tasks and domains, ensuring their effectiveness and relevance across a wide range of applications. In the context of continual instruction tuning, where models are sequentially trained on different tasks, catastrophic forgetting can occur, leading to performance degradation on previously learned tasks. This work addresses the catastrophic forgetting in continual instruction learning through a switching mechanism for routing computations to parameter-efficient tuned expert models. We demonstrate the effectiveness of our method through experiments on continual instruction tuning of different natural language generation tasks and vision-language tasks. We also showcase the advantages of our proposed method in terms of efficiency, scalability, portability, and privacy preservation.**

***Index Terms*—large language models, continual instruction tuning, catastrophic forgetting**

I. Introduction

Large AI models including large language models (LLMs) and multimodal models (MMs) have demonstrated remarkable capabilities across numerous domains, as highlighted by [1] and [2]. However, whereas both LLMs and MMs pre-trained on extensive data excel in general language understanding and multimodal tasks, they may not be optimized for every specific task of interest prompted by instructions. Therefore, there is a need for continual instruction learning to adapt LLMs and MMs to evolving tasks and domains. Indeed, continual instruction learning is essential for LLMs such as GPT [3] and MMs such as LLaVA [4], [5] to maintain their effectiveness and relevance in handling a wide range of tasks and domains.

Such models are trained on vast amounts of data and fine-tuned for specific applications, often by learning tasks sequentially [6], i.e. learning on datasets pertaining to one task all at once, before moving on to the next task. The challenge lies in their ability to continually learn and adapt as they encounter new tasks and information. However, in continual instruction learning scenarios, where models are sequentially trained on different tasks or datasets, catastrophic forgetting occurs when the model's parameters are updated to accommodate new information, leading to degradation or complete loss of performance on previously learned tasks.

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A typical way to balance new learning with the retention of previously acquired capabilities in large models is through replaying old data. However, with rapid iterations of large models for diverse and complex use cases, retaining old data becomes exceptionally challenging. Moreover, continually tuning a model with a large number of parameters is highly costly in terms of both computation and memory usage. Parameter-efficient fine-tuning (PEFT) such as low-rank adaptation (LoRA) [7] provides a lightweight option with portable parameters, which could be paired with a model to perform specific tasks. Therefore, in this work, we focus on alleviating catastrophic forgetting during continual instruction tuning, particularly with minimal data retention and its interplay with PEFT.

We propose a novel continual instruction tuning method, SwitchCIT, that alleviates forgetting of previously seen tasks by introducing a switch network to identify a task for a user query, leveraging a task-specific clustering phenomenon previously demonstrated by [8]. For each new task, we fine-tune the task performance by including extra parameters created by PEFT methods such as LoRA (a self-expansion process by increasing experts) in addition to a pre-trained large base model to build an expert model. Our method shares some similarities with mixture of experts (MoE) [9], but key differences exist: Instead of assigning experts for optimizing a certain objective, SwitchCIT uses distinct models to address different parts of a task space defined by instructions. Additionally, we decouple the learning processes for model selection and the models themselves, whereas MoE jointly learns both.

Catastrophic forgetting in neural networks is related to the palimpsest phenomenon that new memories rapidly overwrite old ones [10], [11]. SwitchCIT may be considered as a way to avoid the need to overwrite the old memories of previously learned tasks by introducing extra parameters for a new task. Moreover, there exists a line of methods inspired by synaptic consolidation in brains that reduces the learning rate on specific weights based on their importance to previously encountered tasks. Our method has the advantage of enabling the full tuning of all weights to adapt to a specific task without restricting any particular set of weights.

We summarize our contributions as follows:

- We propose a novel continual instruction-tuning approach to alleviate catastrophic forgetting by using a switch network inspired by a task-specific clustering phenomenon within LLMs for task routing to different specialized models tuned via PEFT.
- We conduct experiments for instruction-tuning on five continual natural language generation tasks and three multimodal vision-language tasks, demonstrating the effectiveness of our method compared to several baselines across different domains.

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- We demonstrate additional advantages of our method, including efficiency, scalability, portability, and privacy preservation.

II. Related Work

Continual Learning and Catastrophic Forgetting. The study of continual learning focuses on developing algorithms that learn from a continuous stream of data, enabling a model to acquire new knowledge while retaining previously learned information without catastrophic forgetting [12]. Catastrophic forgetting happens when LLMs forget previously learned information as new tasks are learned [6]. In [13], there is an insightful study empirically showing that pre-trained LLMs may forget domain knowledge and tasks that were not included in the fine-tuning process, while supervised fine-tuning offers substantial benefits to the models. To counter this effect, here we use a switch network to classify tasks and route computations from their instructions. By doing so, we can fine-tune task performance by including extra parameters created by PEFT methods such as LoRA for each task.

Continual learning methods are often classified into the following categories: replay-based methods [14], [15], regularization-based methods [16], [17], and architecture-based methods [18], [19]. Catastrophic forgetting arises when model parameters are updated to account for new data, causing degradation or a complete loss of performance on previously learned tasks. Catastrophic forgetting is not specific to LLMs. Other neural networks also experience this phenomenon, leading to methods such as elastic weight consolidation [20]. Previous research to resolve this problem has scaled the number of parameters in the model [21], [22]. It has been demonstrated that these solutions work in theory but suffer from overreliance on scaling LLM parameters. Works such as [16] avoid this by splitting parameters into two sets: one for tasks learned and one to dynamically be generated.

Instruction Tuning. Instruction tuning is the process of tuning a model from specific instructions or prompts that will guide the model toward behaving in the desired fashion. A major issue with LLMs has been the mismatch between the training objective of the LLM and users' objectives. Instruction tuning has been developed, in part, to address this issue. This method of training aims to align language models with human intent [23]–[25]. Moreover, instruction tuning has proven to be highly effective in multimodal settings. [4], [5], [26]. We concentrate our work on the specific case of continual instruction tuning across different tasks, which presents unique challenges such as catastrophic forgetting. Models trained by instruction tuning have been shown to have better performance on unseen tasks than without [27]. Instruction tuning, however, has its challenges of crafting high-quality instructions that cover the desired behavior, and it seems to only capture surface-level patterns rather than truly comprehending the task.

Parameter-Efficient Fine-Tuning. PEFT addresses the challenge of needing enormous computing resources to fine-tune contemporary LLMs. PEFT reduces the number of fine-tuning parameters and memory usage while still achieving similar results as full fine-tuning [28]. One particularly popular PEFT method is LoRA, which freezes the model weights of the pre-trained model and injects trainable rank decomposition matrices into each layer of the Transformer architecture, allowing training on a small number of additional parameters rather than on the original pre-trained model [16]. Here, we use LoRA to create extra parameters for fine-tuning tasks not yet seen by the LLM. Since the original proposal of LoRA, there have been many variants that aim to resolve certain limitations of the original LoRA method, see [29]–[32]. Notwithstanding, LoRA still seems to be the most widely used PEFT method.

Mixture of Experts. MoE models integrate multiple sub-models, or experts, to address different parts of the input space [9], [33], [34]. Though there are some similarities between MoE and our SwitchCIT approach, we use different models to handle different parts of a task space represented by instructions, rather than an arbitrary input space as in MoE models. Also, we separate the learning processes for model selection and the models themselves, whereas MoE models learn both simultaneously. SwitchCIT can self-expand its parameters to adapt to new tasks, whereas MoE models typically do not.

III. Preliminaries

Consider a model parameterized by θ, which is trained to optimize a loss function $\mathcal{L}$ while incorporating human-provided instructions $\{I_t\}_{t=1}^{T}$ over a sequence of tasks T.

The objective of continual instruction tuning is to update the model parameters θ in response to new data received during the learning process. The formulation involves iteratively adapting the model's behavior to align with the evolving set of instruction-based data.

Let us assume there are N_t data examples for a task t: $\{(I_t, x_{t,n}, y_{t,n})\}_{n=1}^{N_t}$, where an instruction I_t, an input $x_{t,n}$, and an output $y_{t,n}$ form a tuple. The continual instruction tuning framework can be defined as follows:

$$\min_{\theta} \sum_{t=1}^{T} \sum_{n=1}^{N_t} \mathcal{L}_t(f_\theta([I_t; x_{t,n}]), y_{t,n}) + \lambda R(\theta)$$

where:

- $[I_t; x_{t,n}]$ is a concatenation of an instruction and an input, where the input $x_{t,n}$ may belong to a different modality, such as vision and is optional,
- $f_\theta([I_t; x_{t,n}])$ denotes the output of the model f parameterized by θ for the task t with an input $x_{t,n}$,
- $y_{t,n}$ represents the target labels or outputs associated with task t,
- $\mathcal{L}$ is the standard loss function measuring the discrepancy between model predictions and true labels for task t,
- $R(\theta)$ is an optional regularization term that penalizes changes to the model parameters θ, helping to mitigate catastrophic forgetting, and
- λ is a hyperparameter that controls the trade-off between optimizing for task-specific performance and preserving knowledge from previous tasks.

In this formulation, the model parameters θ are updated sequentially across tasks $t = 1, \ldots, T$, where each iteration

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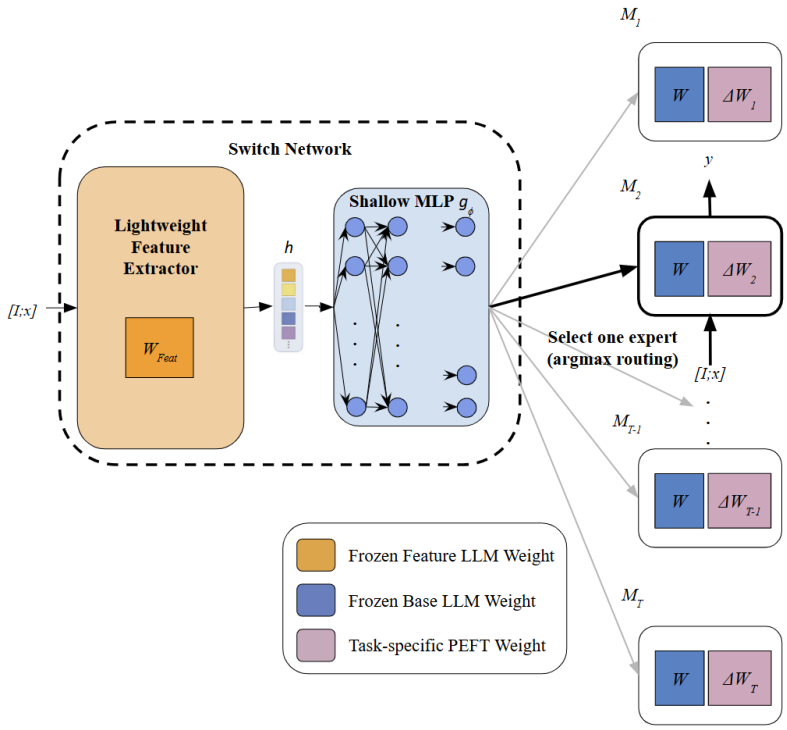


Fig. 1: Given a query consisting of an instruction and optional input $[I; x]$, a lightweight LLM feature extractor with frozen parameters W_{Feat} produces a compact feature representation h. This feature vector h is passed to a shallow MLP network g_ϕ that performs task classification and selects a single expert via argmax routing. Each expert model M_t shares the same frozen base LLM weights W and differs only in its task-specific parameter-efficient adaptation ΔW_t (e.g., LoRA). The selected expert processes the query to generate the output y. By isolating task identification from task execution and routing each query to a dedicated expert, SwitchCIT enables continual instruction tuning while mitigating catastrophic forgetting and avoiding parameter interference across tasks.

involves minimizing a task-specific loss. The continual instruction tuning objective aims to update the model's parameters θ incrementally across tasks, optimizing performance on new tasks while maintaining competence on previously learned tasks. This formulation embodies the principle of lifelong learning, enabling the model to adapt and improve over time without sacrificing past knowledge. Many classical methods, such as [20] rely on the regularization term $R(\theta)$ to alleviate catastrophic forgetting. However, our proposed method approaches the catastrophic forgetting problem differently, without relying on regularization, as will be discussed in the following section.

IV. Method

We illustrate SwitchCIT in Figure 1. Continual instruction tuning proceeds over a sequence of tasks $\mathcal{T} = \{1, \ldots, T\}$. At inference time, a query consisting of an instruction and an optional input, denoted by $[I; x]$, is provided to the system. Based on this query, a switch network routes the computation to an expert model trained explicitly for the predicted task, enabling strong performance on both previously learned and newly introduced tasks.

More specifically, the switch network performs task identification via multi-class classification. The query $[I; x]$ is first encoded by a lightweight language model with weights W_{Feat}, from which we extract the final-layer representation of the last token as a feature vector h. This design is motivated by prior observations that queries corresponding to the same task tend to form well-separated clusters in the hidden representation space of LLMs, with task-specific clustering becoming more pronounced in later layers [8]. Effective clustering in this representation space implies good separability of task identities, which the switch network exploits for reliable routing.

The feature vector h is passed to a multi-layer perceptron classifier g_ϕ, producing a categorical distribution over tasks:

$$p_\phi(t \mid I, x) = \text{Softmax}(g_\phi(h)).$$

An argmax routing mechanism is then applied to select the most likely task,

$$\hat{t} = \arg\max_{t \in \mathcal{T}} p_\phi(t \mid I, x).$$

The query is subsequently routed to the corresponding expert model $M_{\hat{t}}$, which generates the output according to

$$y = M_{\hat{t}}([I; x]) = M_{W + \Delta W_{\hat{t}}}([I; x]). \tag{1}$$

Each expert model consists of a frozen base LLM with parameters W, augmented by task-specific parameter-efficient updates $\Delta W_{\hat{t}}$ (e.g., LoRA). The expert operates on the concatenation of the instruction and input $[I; x]$ and produces an output sequence y, which in this work corresponds to a textual response.

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By routing each query to a dedicated expert, SwitchCIT ensures that models tailored to different tasks do not interfere with one another, thereby alleviating catastrophic forgetting of previously learned tasks. All expert models M_1–M_T are instruction-tuned independently for their respective tasks by introducing task-specific parameters ΔW through a PEFT method such as LoRA. The switch network itself is lightweight and can be implemented using a simple MLP on top of an LLM-based feature encoder. As continual learning progresses, the system self-expands by training a new task-specific expert whenever a new task is encountered.

A recent study [35] attributes forgetting from fine-tuning a large model to the learned bias of its alignment toward a specific task, but without fundamentally altering the model's transferable knowledge across tasks. Our proposed method is designed to align with this perspective: we fine-tune separate models for distinct tasks to prevent potential forgetting of prior task alignments and avoid conflicts among alignments for different tasks. While a limitation of this design is its ineffectiveness in leveraging transferable knowledge across tasks, this concern is alleviated in the era of large models; we expect most essential transferable knowledge to be acquired during the pre-training stage, reducing the need for further obtaining it during continual fine-tuning. Another limitation of our method is its reliance on good task separability. We anticipate that future advancements in text encoders will enhance the ability to capture finer distinctions between tasks, thereby improving our method's effectiveness in this regard.

V. Experiments on Language Tasks

In this section, we briefly overview the model implementation and datasets. Experimental setups are further detailed below.

We use BLOOMZ 1.1B, BLOOMZ 7.1B [36] and Llama-3.1-8B [37] as choices of our base LLMs and LoRA to learn task-specific and portable parameters [7]. A switch network is implemented using a two-layer MLP network. We use OPT-125M, a lightweight LLM with only 125 million parameters to extract features for the switch network [38]. Implementation and training details are in Section V-A.

Choosing BLOOMZ LLMs as our base models allows us to use the benchmark introduced by [39] and five continual instruction tasks for natural language generation selected by [6] to differ from the training and evaluation tasks of BLOOMZ to prevent data leaking. These tasks are:

- Text Simplification (Simp), paraphrasing a given text into simpler language [40], [41];
- Empathetic Dialogue Generation (Emdg), generating a dialogue response that offers a reason within a specified emotional context [42];
- Inquisitive Question Generation (InqQG), creating questions that prompt long-form answers [43];
- Explanation Generation (Exp), generating natural language explanations for provided premises, hypotheses, or labels [44]; and
- Headline Generation with Constraint (HGen), producing headlines that include specific keywords, positioned as per the constraints specified [39].

We evaluate a model's performance on each task using metrics outlined by [39]. Specifically, we use SARI for Simp; BERTSore (BS) for Emdg, InqQA, and Exp; and Rouge-1 (R1) for HGen. We use the BERTScore version based on DeBERTa-MNLI.

Following [6], we train a model based on the order of instruction tasks: Simp $\rightarrow$ Emdg $\rightarrow$ InqQG $\rightarrow$ Exp $\rightarrow$ HGen. We use the specific prompts designed by [39] and train on 100,000 data samples, following previous works [6], [39]. We feed an instruction possible with an optional input to the switch network for determining task identity since in practice, an instruction and an input may not be easy to separate.

We compare our method to a direct supervised fine-tuning approach and a continual learning method with a rehearsal mechanism following [39]. We consider the rehearsal method because it is the only method in the literature that is developed in an experimental setting closest to our continual instruction tuning setting and a representative of continual learning methods based on data replaying. Both our method and the other methods employ the same PEFT using LoRA unless specified otherwise.

To ensure a fair comparison, we use the same amount of data for rehearsal as we do for training our switch network. To investigate the regime of low data retraining, we reserve only 0.01% of the training data for each task to train the switch network, which is significantly less than the amount of replay data such as the 1% used by traditional continual learning methods for rehearsal. We also evaluate methods using 1% of the training data, based on which the rehearsal method was originally developed.

We additionally compare our approach with a Mixture-of-Experts (MoE) baseline by performing supervised instruction tuning on LLaMA-MoE-v1-3.0B-2/16 [45], which activates 2 out of 16 experts (for a total of 3.0B parameters), following the same task sequence described in Section V. We select this model because its authors also provide a dense counterpart, Sheared-LLaMA-2.7B, with a comparable parameter count and performance, enabling a fair comparison when applying our method.

We also compare against a multi-branch baseline. Inspired by prior work in other domains [46], we construct a multi-branch model with a prototype-based routing mechanism. Specifically, routing is determined by the maximum cosine similarity between the embedding of the input query and a task prototype. Embeddings for examples are computed using the same lightweight LLM employed by our switch network. We consider two variants for constructing task prototypes: (1) encoding a task description, and (2) averaging the embeddings of training examples associated with each task. This design enables a direct comparison, as the routing strategy is the only differing component.

A. Implementation Details

During instruction tuning, we follow [39] to start by adding a general prompt template at the beginning of the data: *'Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately*

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TABLE I: Hyperparameters related to switch network and its training.

Hyperparameter	Value
Learning rate	1×10^{-3}
Number of epochs	20
Optimizer	AdamW
Scheduler	Constant
Number of layers	2
Hidden dimension	200

TABLE II: Hyperparameters related to language generation models undergoing continual instruction tuning and their training. We adopt the training hyperparameters from [6].

Hyperparameter	Value
Learning rate	2×10^{-5}
Number of epochs	3 per task
Optimizer	AdamW
Scheduler	Constant
LoRA rank	64
LoRA Alpha	16
LoRA dropout	0.05
LoRA bias	None

completes the request...'. This is then followed by a specific prompt for each task.

We implement our switch network via OPT-125M [38] as a feature extractor and its classifier using a two-layer MLP with ReLU activation function [47].The switch network is retrained after each new task using both the retained data from the new and previous tasks. During training, the base model is kept frozen and only the LoRA weights are updated. We present hyperparameters related to the switch network in Table I and to continual learned models in Table II.

We perform the experiments on 2 X A100 Nvidia GPUs with 80 GB GPU memory.

B. Switch Network

Table III presents the progressive performance of task classification by our switch networks trained under different conditions: a low data retention rate of 0.01% and a rate of 1% comparable to the replay data used by the rehearsal method of [39]. Note that after learning each task, we retrain a very lightweight task classifier to accommodate the newly learned task. It is evident that switch networks of different settings achieve very high classification accuracy at every learning stage, even when using a lightweight LLM like the OPT-125M for feature extraction. The performance is scaled up by using more training data. Performance does not obviously degrade when including more tasks, demonstrating good robustness.

TABLE III: Progressive performance of task classification by our switch networks measured by accuracy. A continual learning stage is denoted by its task name. Note that there is no switch network for the first learned task.

Retention Rate	Emdg	InqQG	Exp	HGen
0.01 %	100.0 %	96.3 %	97.6 %	97.1 %
1.0 %	100.0 %	99.9 %	99.9 %	99.6 %

Notice that competitive classification performance is reached even with 100× less data, which may be explained by good separability of different tasks due to task clustering shown by [8].

C. Analysis of Switch Network

To study robustness of our method toward tasks that are not well-separated, we introduced an additional task, summarization, which is similar to simplification, to evaluate how our model handles similar yet distinct tasks. This task included 16 training examples and 25 held-out test examples, with instructions for summarization differing from those used in the training data. Our evaluation shows that the model trained on this additional task achieved 95.52% accuracy overall and 90.23% for the summarization task, demonstrating its ability to reliably distinguish between two similar tasks after training. Overall, for similar tasks that lack clear separability, we merge them into a single routing class and handle them with a shared expert, since each expert can perform unconstrained free-form generation. Then, the expert can learn transferable skills across these tasks.

Treating our existing test data as in-distribution (ID), we constructed a comprehensive out-of-distribution (OOD) test set. The OOD set consists of five categories: 75 ambiguous examples (overlapping and ambiguous instructions), 59 out-of-domain examples (tasks from different domains like code and math), 34 nonsensical examples (impossible or nonsensical instructions), 19 conversational examples (open-ended conversational queries), and 8 command examples (action commands for virtual assistants).

We then compare the confidence levels of our method on both the in-distribution and out-of-distribution data. We evaluate model confidence using several metrics:

- **Max Probability:** The highest probability after applying softmax. Values close to 1.0 indicate high confidence, while values near 0.5 suggest uncertainty. This metric only considers the top predicted class.
- **Margin:** The difference between the top two predicted probabilities. A large margin indicates a clear preference, whereas a small margin reflects ambiguity useful for detecting uncertainty between competing predictions.
- **Entropy:** Measures the uncertainty across the entire probability distribution. Higher entropy indicates greater uncertainty, while lower entropy suggests more confident predictions. Unlike Max Probability or Margin, it captures how spread out the probabilities are.
- **Energy [48]:** Computed using raw logits as the negative log-sum-exp. Lower energy values correspond to higher confidence.

As shown in Table IV, our method exhibits significantly higher average confidence on ID data compared to OOD data across different metrics, demonstrating its ability to effectively distinguish between the two. When the model detects low confidence in an OOD example, we employ a fallback mechanism that routes the computation to the base LLM.

A straightforward approach to OOD detection is to apply a threshold-based decision rule, where a scalar confidence or

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TABLE IV: Confidence and uncertainty metrics of our switch network on in-distribution (ID) and out-of-distribution (OOD) data. Gap denotes the absolute difference between OOD and ID means. Arrows indicate the direction associated with higher confidence.

Metric	ID (Mean $\pm$ Std)	OOD (Mean $\pm$ Std)	Gap $\lvert$OOD $-$ ID$\rvert$
Max Probability $\uparrow$	0.982 ± 0.082	0.768 ± 0.200	0.213
Margin $\uparrow$	0.971 ± 0.131	0.616 ± 0.330	0.354
Entropy $\downarrow$	0.055 ± 0.197	0.605 ± 0.435	0.550
Energy $\downarrow$	-7.703 ± 2.519	-3.529 ± 2.256	4.174

TABLE V: OOD detection performance comparison across different routing methods. AUROC is reported together with the absolute energy gap between OOD and ID means. Higher is better for both metrics.

Method	AUROC $\uparrow$	Energy Gap $\lvert$OOD $-$ ID$\rvert$ $\uparrow$
Task Prototype (description)	0.635	0.055
Task Prototype (average)	0.639	0.058
Switch Network	**0.831**	**4.174**

score (e.g., energy, maximum probability, or margin) is compared against a predefined threshold to determine whether an input should be treated as in-distribution or out-of-distribution. Inputs exceeding (or falling below, depending on the metric) this threshold are flagged as OOD, while those within the threshold are considered in-distribution.

To evaluate this, we constructed a balanced OOD detection dataset consisting of 195 OOD examples and 195 randomly sampled ID examples, evenly drawn from each task. Using energy as the confidence metric, we compare threshold-based OOD detection performances of different methods using their AUROC. Table V shows that our method consistently outperforms the two task prototyping methods in AUROC and exhibits a substantially larger energy gap between in-distribution and out-of-distribution examples, indicating stronger distributional separation and our method's advantage on OOD detection.

D. Continual Instruction Tuning

The results presented in Table VI are derived from experiments conducted with the base LLM, BLOOMZ-1.1B, retaining 1% training data. As shown in Table VI, the performance of the model continually tuned with PEFT degrades significantly on previously-learned tasks compared to the initial model, as expected. However, our SwitchCIT method demonstrates substantial improvements over the initial model across all tasks. This indicates that SwitchCIT effectively mitigates forgetting and reaps the benefits of continual instruction tuning. Notably, SwitchCIT outperforms the PEFT model on the HGen task, which is the last task tuned, suggesting that our approach may effectively reduce negative interference from prior tasks. Furthermore, SwitchCIT exceeds the performance of the rehearsal method that uses the same retention data rate (1%) as studied by [39]. This comparison highlights that our method utilizes the retained data more effectively than rehearsing previous data.

We present studies using base LLMs of varying scales and types and different data retention rates in Table VII. We compare SwitchCIT (BLOOMZ-1.1B and 0.01% data retention) relying on PEFT to a much larger fully supervised fine-tuned approach based on the BLOOMZ-7.1b model [6], [36], which is approximately seven times larger than our base LLM. Surprisingly, our method still achieves better performance on most tasks, underscoring its efficiency and effectiveness. We also include results using the larger base LLM, BLOOMZ-7.1B, which achieves better performance with 1% data retention, demonstrating the scalability of our method. In contrast to the rehearsal method, our approach experiences notably less performance degradation when retaining 100 times less data from 1% to 0.01%. This performance gap across most tasks also remains minimal for our SwitchCIT, when using the larger base LLM, BLOOMZ-7.1B. Furthermore, we observe the same trend with LLaMA 3.1-8B at a 0.01% data retention rate, indicating that forgetting remains a significant issue even in a larger and more recent LLM. This suggests that our method is effective not only with BLOOMZ models but also with more advanced models like LLaMA 3.1-8B.

We further conducted evaluations on both an MoE model and a multi-branch model based on a routing mechanism via task prototyping.

For the experiments involving MoE models, as shown in Table VII, our approach, using the Sheared-LLaMA-2.7B base model, consistently outperforms the MoE baseline across all tasks after the learning sequence, with particularly stronger performance on previously learned tasks.

Regarding the comparison to multi-branch models, based on Table VII, our method consistently outperforms two variants of this prototype-based multi-branch model on both retaining performance on previously learned tasks and generalizing to new ones.

We hypothesize that due to our high-performing switch network, our method is able to specifically select a tailored sub-model for a task, yielding impressive performance. This also suggests that SwitchCIT requires much less data retention compared to the rehearsal method, highlighting the practicality and efficiency of our approach.

To evaluate the success of the continual instruction tuning over a sequence of N tasks, we use the concept of upper bound (UB) by following [39]. The UB is the highest performance that the model can achieve when fine-tuned exclusively on a specific task, T_n. The model is considered to be successful if it maintains a performance close to the UB while learning new tasks. The normalized results, referred to as relative gain (RG) for a given task T_n, are calculated as:

$$RG_{T_n} = \frac{s_{T_n}}{UB_{T_n}} \tag{2}$$

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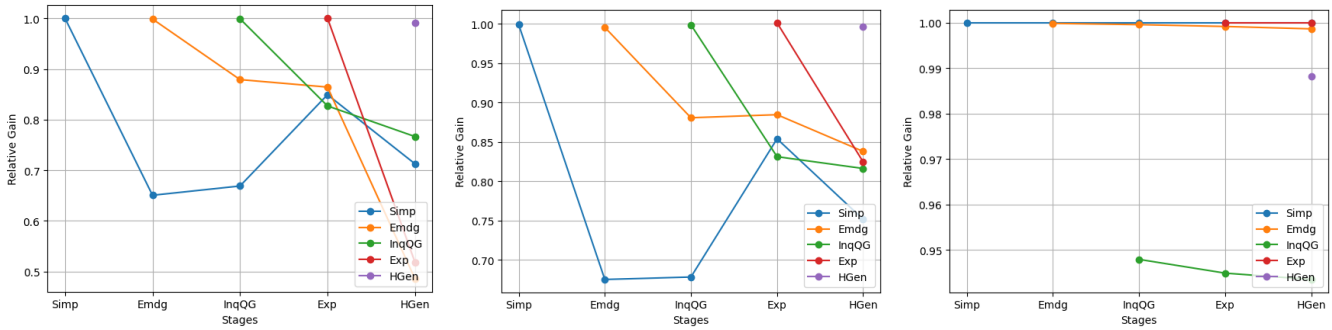
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TABLE VI: The final performance of various methods on different tasks in the continual instruction tuning. Tasks are presented in the learning order. We also list the performance of the original LLM as "Initial".

Method	Simp	Emdg	InqQG	Exp	HGen
Initial	37.7	0.483	0.457	0.515	0.269
PEFT	34.9	0.265	0.454	0.369	0.356
Rehearsal	48.4	0.533	0.589	0.685	0.357
SwitchCIT	**49.0**	**0.546**	**0.593**	**0.712**	**0.359**

TABLE VII: The final performance of various methods in different settings on different tasks in the continual instruction tuning. Tasks are presented in the learning order. Full-SFT refers to the fully supervised fine-tuning method without using the PEFT.

Method	Base LLM	Retention Rate	Simp	Emdg	InqQG	Exp	HGen
Full-SFT	BLOOMZ-7.1B		47.2	0.533	0.597	0.687	0.329
Rehearsal	BLOOMZ-1.1B	0.01%	36.8	0.458	0.484	0.587	0.359
Rehearsal	Llama-3.1-8B	0.01%	40.5	0.460	0.567	0.621	0.404
Rehearsal	BLOOMZ-1.1B	1%	48.4	0.533	0.589	0.685	0.357
SwitchCIT	BLOOMZ-1.1B	0.01%	49.0	0.545	0.559	0.712	0.355
SwitchCIT	BLOOMZ-7.1B	0.01%	49.5	0.561	0.577	0.726	0.414
SwitchCIT	Llama-3.1-8B	0.01%	48.6	0.561	**0.636**	0.712	0.395
SwitchCIT	BLOOMZ-1.1B	1%	49.0	0.546	0.593	0.712	0.359
SwitchCIT	BLOOMZ-7.1B	1%	49.5	0.562	0.615	0.726	**0.418**
MoE-SFT	LLaMA-MoE-v1-3_0B-2_16		34.2	0.446	0.472	0.520	0.397
Task Prototype (description)	Sheared-LLaMA-2.7B	0.01%	50.14	0.4791	0.5616	0.5832	0.2119
Task Prototype (averaging)	Sheared-LLaMA-2.7B	0.01%	41.92	0.5619	0.5465	0.7264	0.3594
SwitchCIT	Sheared-LLaMA-2.7B	0.01%	**50.2**	**0.564**	0.596	**0.744**	0.399



(a) PEFT (b) Rehearsal (c) SwitchCIT

Fig. 2: Progressive relative gain of various methods. The horizontal axis presents different learning stages labeled by their respective task names, whereas the vertical axis shows the relative gain. Task performances are shown once the task is learned across different stages.

where s_{T_n} is the performance of a continually-tuned model on task T_n. A high relative gain indicates effective retention of performance on a specific task and a value of 1 indicates that the model performs at the UB for a task.

From Figure 2, notice that SwitchCIT based on BLOOMZ-1.1B at 1% data retention rate experiences minimal catastrophic forgetting compared to other approaches, as shown by almost perfect retention of performance for various tasks. The less perfect retention on the InqQG task is due to imperfect task identification by the switch network. In contrast to our method, the rehearsal method retains only some performance on previous tasks. Along with PEFT, they both exhibit noticeable levels of catastrophic forgetting as they continue to learn additional tasks.

VI. Experiments on Multimodal Tasks

Multimodal tasks involve processing and understanding information from multiple data modalities (e.g., text, images, audio, video, structured data). These tasks leverage the complementary strengths of each modality to enhance performance and enable capabilities that are otherwise impossible with a single modality.

Models for handling vision-language tasks have been widely studied. Instruction tuning is a crucial step for vision-language models (VLMs) because it allows them to generalize better, align more closely with user intentions, and handle a wide variety of tasks by following explicit natural language instructions. However, current VLMs are typically trained in a multitask setting, and continual instruction tuning for VLMs remains largely unexplored. This is becoming increasingly demanding as the vision-language tasks a system is expected to handle are becoming more complex and diverse.

We focused our experiments on a recent and widely adopted vision-language model (VLM), LLaVA v1.5, which has demonstrated impressive performance on vision-language tasks in a multitask setting with high training efficiency [5]. Its training

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TABLE VIII: The final performance of various methods on different vision-language tasks in the continual instruction tuning. Tasks are presented in the learning order.

Method	VR	OCR-VQA	IMG-CAP
Rehearsal	0.0	0.150	**1.347**
SwitchCIT	**0.593**	**0.545**	1.310

pipeline includes a pre-training stage followed by an instruction-tuning stage. In this work, we decompose its original multitask instruction tuning stage into a continual instruction tuning of individual tasks.

We study three distinct tasks: Visual Reasoning (VR), OCR-based Visual Question Answering (OCR-VQA) and Image Captioning (Img-Cap), as they are sufficiently different and have well-established benchmarks. For the VR task, we utilize the GQA dataset for both training and evaluation [49] by following the setup from the LLaVA v1.5 [5]. For OCR-VQA, we perform training on the OCR-VQA dataset [50] and evaluation on the scene text-centric VQA data from the OCRBench benchmark [51]. We used the COCO image captioning dataset [52], [53] for both training and evaluation (the 2017 validation split) of the Img-Cap task. Task performances on both VR and OCR-VQA are reported in terms of accuracy, while we report CIDEr [54] score for Img-Cap. We adopt a continual learning order of GQA $\rightarrow$ OCR-VQA $\rightarrow$ Img-Cap.

Our approach follows the parameter-efficient instruction tuning strategy with LoRA proposed by [5]. Specifically, we use Vicuna 7B v1.5 [5] as the base LLM and employ the same switch network setup described in Section V. For comparison, we include a rehearsal-based baseline that relies on 0.01% data replay.

Table VIII and Figure 3b compare our method to the rehearsal-based baseline. The results reveal that even with data replay, the VLM exhibits significantly worse forgetting of previously learned tasks compared to its LLM counterparts. We hypothesize that this is because VLMs often receive similar visual inputs across tasks, making it difficult to distinguish among them. In contrast, our SwitchCIT method effectively mitigates the forgetting issue, as evidenced by the high relative gain shown in Figure 3b. This demonstrates not only the effectiveness of our approach but also its robustness across different application domains.

VII. Additional Advantages

In this section, we discuss additional advantages of our method beyond its superior performance.

A. Efficiency

Many existing works overcome catastrophic forgetting by imposing constraints on the existing parameters of a model, so their model sizes do not change. SwitchCIT introduces a small number of new parameters for each additional task. For example, when using BLOOMZ 1.1B as the base LLM, these additional parameters account for only 0.878% of the total parameters. Considering five continual tasks, the additional parameters amount to just 4.39% in exchange for minimal catastrophic forgetting, demonstrating their lightweight and practical feasibility. Note that only the additional parameters specific to a single task are loaded during inference. We anticipate further improvements in these numbers as parameter-efficient methods continue to advance.

B. Scalability

The size of a multitask learner usually needs to scale with the number of tasks to maintain high performance across all tasks. However, this scaling can lead to increased response latency as the number of tasks grows. In contrast, our method enables the creation of a constant-size model tailored to new tasks while maintaining strong overall performance. This ensures that latency is minimally affected as the number of tasks increases; the lightweight switch network may scale slowly with the number of tasks, but the resulting latency increase remains minimal. This is particularly crucial as we expect systems to become increasingly versatile in the future, yet minimizing latency is essential for scenarios where timely responses are critical, such as autonomous driving and financial trading.

Conceptually, SwitchCIT is designed to be scale-compatible with very large LLMs because it operates purely through low-rank adapters and a lightweight routing mechanism on top of a frozen backbone. At a very large scale, the main bottlenecks are (i) GPU memory for storing many adapters, and (ii) integrating routing and adapters parameters into tensor- or pipeline-parallel inference. We can have mitigation strategies: for example, sharding adapter sets across devices, or using more aggressive rank compression for high-capacity models.

C. Portability

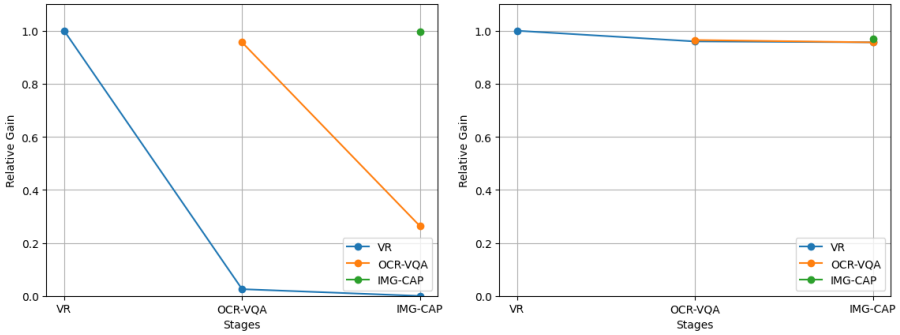
Separating the development of the switch network from the instruction-tuned models greatly enhances SwitchCIT's portability. For instance, to improve task identification by our switch network using more data (from 0.01% to 1.0%, as shown in Table III), we only need to retrain the switch network and plug it in the existing instruction-tuned models. Conversely, we can also use existing switch networks for better instruction-tuned models as shown in Table VII, where we leverage the same switch network for models with larger base LLMs such as BLOOMZ 7.1B and Llama 3.1-8B.

D. Privacy

Privacy is a crucial consideration as the data used to train, test, and deploy models often includes sensitive, personal, or proprietary information. Ensuring privacy not only safeguards individual rights and organizational interests but also promotes ethical, legal, and trust-related aspects of AI systems.

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(a) Rehearsal (b) SwitchCIT

Fig. 3: Progressive relative gain of various methods. The horizontal axis presents different learning stages labeled by their respective task names, whereas the vertical axis shows the relative gain. Task performances are shown once the task is learned across different stages.

Our parameter-efficient task-specific models can be trained separately in a distributed manner, offering a robust privacy-preserving capability. For instance, sensitive data, such as medical records, can remain localized for model training, while the trained model can still be utilized during inference. Privacy-preserving features could potentially enable collaborations among different industries, where one party provides the technology and another contributes application-critical but sensitive data.

Our method is somewhat akin to federated learning. However, federated learning [55] primarily aims to train a single global model by exchanging gradients or parameter updates across many rounds. Even though data remains on-device, the global backbone is continuously updated, and gradient-sharing is known to be vulnerable to information leakage (e.g., reconstruction or membership inference attacks). In contrast, SwitchCIT keeps the backbone fixed and introduces skill-wise LoRA modules and routing as the locus of adaptation. New capabilities can be added or revoked by attaching/detaching small skill adapters, without requiring iterative gradient exchange or global model updates. Thus, our privacy advantage is architectural: we reduce the attack surface to a modular, low-rank interface rather than a constantly updated full model.

Besides, unlike adapter-sharing methods [56] that pool adapters and apply them uniformly across tasks, SwitchCIT includes an explicit routing mechanism that conditions on which skill adapters are activated on the intended instruction. This intent-aware routing could yield a finer-grained separation of skills and potentially avoid the need to expose or merge adapters that correspond to sensitive or high-risk instructions.

VIII. Limitations

Because of computational constraints, we could only test our method on models of limited scale. However, according to our design, our high-performing switch network is independent of the base models and can be paired with larger-scale models that offer superior performance. The remarkable performance of our switch network showcases the effectiveness of our method. It is worth noting that our task classifier in the switch network is incredibly lightweight (154K parameters) and requires minimal data (0.01% of the training data), making it highly practical and easy to integrate.

IX. Conclusion and Discussion

We proposed a novel continual instruction-tuning approach to alleviate catastrophic forgetting by using a switch network to identify tasks by leveraging the task-specific clustering phenomenon and then route computations to parameter-efficient tuned expert models. Experiments conducted on five instruction-based continual natural language generation tasks and three continual vision-language tasks demonstrate the effectiveness of our method across different domains by comparing it to several baselines. We also analyze and show the efficiency, scalability, portability, and privacy preservation of our proposed method. Note that our method is broadly applicable to the standard pre-training and fine-tuning framework for large models by treating these two stages as two sequentially learned tasks and addressing the issue of potential forgetting accordingly. However, how to view pre-training and fine-tuning as two distinct tasks is to be carefully defined based on specific use cases, ensuring that the switch network can reliably identify the appropriate task for a given user query.

Our proposed method has the potential to bring positive societal impacts, including better collaboration across industries due to its privacy-preserving properties and environmental benefits resulting from its improved efficiency.

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